

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE
OR BRAND NAME

PICK ONE

Sole Source:

☒ I recommend that NASA, NASA Shared Services Center negotiate with AME only for INRODES Phase 1 RDRE Hardware. The total quoted cost of this effort is [REDACTED] (two hundred and forty eight thousand and nine hundred dollars) and the estimated period of performance or lead-time for delivery is June 1, 2025.

OR

Brand Name Justification:

☐ I recommend that NASA, NASA Shared Services Center negotiate with only with vendors who can provide the subject brand name from Manufacture Name for Enter Supply or Service to be procured. The total estimated cost of this effort is Enter Dollar amount and the estimated period of performance or lead-time for delivery is Date.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacture. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

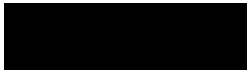
The primary goal of this INRODES project award is to partner with an additive manufacturing (AM) supplier that is able to explicitly meet the requirements for powder bed fusion components laid out by the project charter. These requirements are as follows. 1.) Provide hardware with no witness lines, restarts or coater arm tears. 2.) Provide all surface features including internal channels with specific surface roughness requirements of 400 micro inches SA ± 25 . 3.) Provide hardware with shrinkage tolerances of $< 0.5\%$ full scale. 4.) Provide hardware with orifice geometries that meet discharge coefficient error of $< 5\%$ compared to predicted values. Finally, several of the key features printed within each set of hardware including integrated coolant channels and injector orifices, are extremely small and represent significant risk to powder blockage. Powder blockage would result in failure of the part and loss in deliverable. Currently, AME has proprietary methods for removing powder from the most extreme integrated features and have demonstrated in past projects (RDRE ECI) that they can successfully remove powder and provide a usable part.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

For the AM laser powder bed fusion process, these requirements are not easily met. A substantial investment in time and money is required to obtain any degree of certainty that these tight error bars can be met. For example, if any other vendor were selected, the error bars would more likely be closer to $\pm 55\%$ error based on experimental data obtained under the RAMPT, ORCA, ALPACA, and RDRE ECI projects. This degree of uncertainty without critically obtained data to scale to would mean the failure of this project and represent a substantial waste in government funds for this project. All hardware would not be usable and the project would fail. If another vendor were selected, a substantial cost overrun of approximately 2.5-4 million dollars would be incurred. This is the approximate investment to date to obtain critical scaling data to achieve the requirements above. Finally, AME utilizes proprietary internal data from previous

component development projects that meet the requirements above that no other vendor within the USA legally has access to. Without these proprietary scaling data, the requirements above will not be met. Finally, the impact on schedule would be significant if AME were not selected. Outside of budget, the data required would cost approximately an additional 3 years in schedule which would out right fail the INRODES efforts. This vendor, AME, is the only vendor capable of supply hardware with the explicit requirements above to meet the project goals. The quote is competitive for AM components and represents excellent cost-benefit to the government.

ADDITIONAL COMMENTS:



Date



Title

Office